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Pathological Myopia Image Analysis Using Deep Learning

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Abstract

Pathological Myopia (PM) is one among the main reason of visual defect in the world. Pathological myopia is associated with decaying changes in the retina. If it remains untreated it may lead to vision loss that can't be recoverable. The correct diagnosis of pathological myopia will facilitate proper treatment and improve disease management which reduce the growth of the disease. However, it is nearly impossible to scan the whole population. Computer-aided diagnosing tools in eye image study will build the method scalable and economical. This paper focuses on the problems of classification of Pathological myopia images and non-pathological myopia images and optical disc, fovea detection, localization, lesions (atrophy and detachment) segmentation with 400 samples provided by International Symposium on Biomedical Imaging (ISBI). In this paper, a deep learning method with Convolutional Neural Networks (CNN) is used for classification and U-net model is used for Image segmentation which shows that it achieves highly competitive results.

Keywords: Pathological Myopia, Medical Image Processing, Deep Learning

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1. Introduction

Pathological myopia is also known as high myopia or degenerative myopia. It is a type of severe and progressive near-sightedness described by distortion in the fundus of the eye. It causes speedy changes in eye vision which often requires a change in eyeglass or contact lens for every four to six months. The condition usually does not stabilize within normal limits thus affects the curvature of the crystalline lens and increases the risk of retinal detachment. In United States, it is identified as the seventh leading cause of blindness in adults. A study stated that 0.5% of the European population are affected by this disease. It is estimated to represent 2% of all types of myopia. The prevalence of this disease increases in Singapore, Hong Kong and Taiwan. Myopia-related visual impairment affects the efficiency and nature of life.

Myopia has become a global burden of public health. Among myopic patients, about 35% have high myopia. Myopia leads to extension of axial length, potentially causing pathological changes in retina and choroid. With an increase in myopic refraction, high myopia will develop into pathologic myopia, which is characterized by formation of pathologic changes at: (1) posterior pole, including tessellated fundus, posterior staphyloma, retino-choroidal degeneration, etc; (2) optic disc, including parapapillary atrophy, tilting, etc.; (3) myopic maculopathy, including lacquer crack, Fuchs spot, etc. Pathologic myopia causes irreversible visual impairment to patients. So, it is important to have timely diagnosis and regular review.

In this paper, a Deep Learning method is used, where Convolutional Neural Network model is used for Classification of Pathological Myopia Images and U-net model is used for Image Segmentation for locating Optic Disk and finding Atrophy lesions.

2. Related Work

Jiang Liu and et al [1] has proposed a method called PAMELA which detects pathological myopia through parapapillary atrophy and naturally evaluates a retinal fundus image of pathological myopia. "Detection of Pathological Myopia by PAMELA with Texture-Based Features through an SVM Approach" [1] work focuses on the texture based features that helps in texture analysis of image component of PAMELA and image data. Support vector machine (SVM) based classification algorithm was used for the detection of the existence of pathological myopia in an image of retinal fundus. They achieved result of 87% on image dataset of Singapore Eye Research Institute. In paper [1] the authors used a machine learning technique, Support Vector Machine (SVM), where the texture based features were extracted from retina fundus images. This may not good as compared to Deep Learning techniques because Convolutional Neural Network (CNN) gives ground-breaking results for Image Processing [2]. CNN obtains abstract features when it propagates towards the deeper levels.

Zhuo Zhang at al [3] has developed a method where apart from retinal images other data they have considered. They were genetic, demographic and clinical information. In their paper they developed "Automatic Diagnosis of Pathological Myopia from Heterogeneous Biomedical Data" [3]. They proposed a computer-aided diagnosis method for the identification of pathological myopia named PM-BMII. It identifies pathological myopia automatically based on the different combination of data sources, i.e. retinal fundus image data, clinical data, and genes related data. Their results with multiple combinations achieved 88% Area under the Curve (AUC). But deep learning technique, Convolutional Neural Network [3], with retinal image processing gives more accurate results to classify Pathological myopia and Non-Pathological Myopia.

The fovea is an important component of the eye. It is located in the center of Macula. Vision loss depends on severity level of retinal diseases, and its abnormalities appear in the eye nearby the fovea location. So, it is one of the important attributes to detect the features of the pathologies related to retina. Fovea is a point where main vision is located. If any changes detected near fovea then importance need to be given. Shilpa Joshi and Dr. P. T. Karule have done the detailed studies in their paper entitled "Fundus Image Analysis for Detection of Fovea: A Review" [4] in which they described the use of segmentation for fovea detection.

It is very difficult to identify Fovea from the picture of fundus with the help of a manual or automatic system. The research is done till the segmentation of vessel network and segmentation of optic disc. The accurate location of the fovea is very hard to detect on fundus images even for an eye experts. High definition of fluorescein angiogram gives clue for the detection of fovea, which highlights the area near fovea. So it is easy and robust to detect fovea since it has geometric relationship with other structures. Based on this study several methods have been developed, but performance evaluation shows that there is still a need of improvement of the result. In this paper, a deep learning technique with U-net model is used for Image Segmentation. U-net model is a Convolutional Network designed for Biomedical Image Segmentation [5] which gives good results when large annotated training sample is available.

Meindert Niemeijerc, Michael D. Abràmoffb and Bram van Ginnekena presented automatic system in "Fast Detection of the Optic Disc and Fovea in Color Fundus Photographs" [6] to locate optic disc and the fovea in retinal fundus. They used kNN-regressor Machine Learning technique and a circular template to estimate the distance to a location in the image. Their method first locates the optic disc in the retina fundus image and then searches the location of fovea based on the optic disc. But the problem is that it is very difficult to find these features in the image. Compare to these techniques Deep Learning U-Net model is very accurate because each layer of convolutional network extracts the high-level features.

Sudharshan et al. [7] presented a new method for the detection of optic disc and fovea in the "Relation Networks for Optic Disc and Fovea Localization in Retinal Images". In their method, they focused on the problem of detection of the centers of the Optic disc and Fovea. Their method localize the centers of the Optic disc and Fovea by parallel processing them and modelling their relative figures and presence. These approach give center location of optical disc and fovea. But, identification of lesions required the whole optical disk segmentations.

3. Proposed work

Research is done on either on detection of Pathological Myopia using machine learning techniques or optic disc and fovea segmentation for minimum datasets. To address all the problems, this paper uses a method called Pathological Myopia Image Analysis using Deep Learning. The architecture of the method is given in below Fig. 1. The method starts from the scratch i.e. from given retina fundus image that belongs to pathological myopia or non-pathological myopia class. It makes an analyses of the given test retina image and predicts the class it belongs to either Pathological Myopia (PM) or Non-pathological Myopia (NM) using deep learning CNN model, and if the retina fundus found pathological myopia then it segments into affected area of lesions (retinal detachment and atrophy) using U-net model.

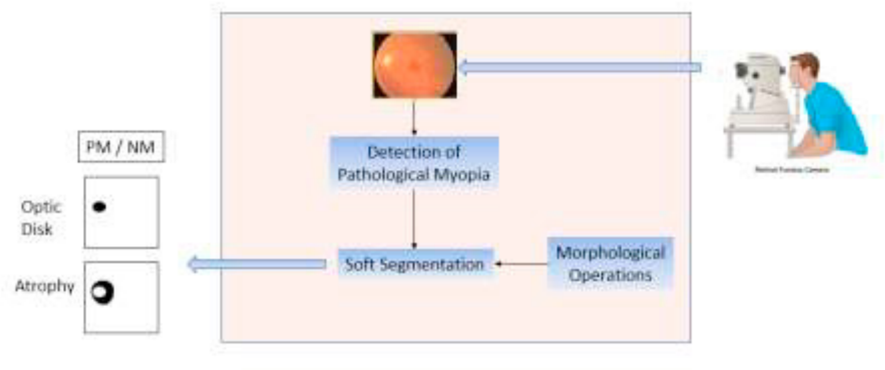

Fig. 1. Workflow Diagram

This proposed method aims to detect pathological myopia from retinal fundus image and segmentation of affected area by various lesions. The dataset consists of 400 color images of 1444 x 1444 size retina fundus in which 237 images belong to PM class and 163 images of NM class. The segmentation dataset contains another 2 sets of annotated images of optical disk and lesions. Annotated image is basically black and white image where region of interest contains black color and rest of the image as white. Dataset is accessed from International Symposium on Biomedical Imaging (ISBI), Italy. Label for Pathological Myopia presence obtained from the health records, which is not based on fundus image ONLY, but also take Optical Coherence Tomography (OCT), Visual Test, and other facts into consideration.

Fig. 2 shows insights of dataset, which is used in this paper. First column is raw retina fundus images of a person. Second column is ground truth value of optical disk region of column 1 images and Third column is ground truth value of atrophy retina lesions. Fourth column is class label for column 1 images. Column 1 and 4 dataset fed into CNN model for the classification of PM and NM, and column 1,2 and 3 fed into U-net model for image segmentation of Optic disc and various lesions in images.

In machine learning, modelling (model training) is an important thing. During the modelling, machine learning model learns from the dataset and fits a mathematical function on it. This process is called training or learning. Generally, in image processing, larger the training set, better the model. In implementation of CNN and U-net deep learning model are used input and an output layer as well as multiple hidden layers.

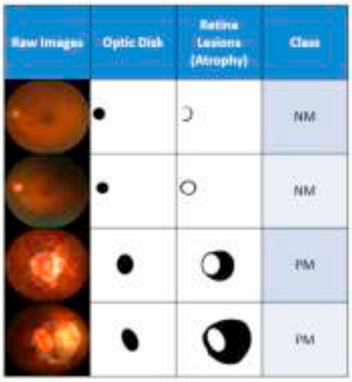

Fig. 2. Dataset insights

The CNN model consists of three convolutional layers and three max pooling layers for image feature extraction. For convolutional layer ReLU activation function is used and 2 dense hidden layers with one SoftMax activation function for final output layer.

The U-net architecture is used for image segmentation. It contains contraction path and expansion path. Contraction path is a Series of twice 3×3 Convolution and 2×2 max pooling. This structure can extract more advanced features but it also decreases the size of features, but it keeps only important features. Expansion path is ensuing of 2×2 Up-convolution and two sets of 3×3 Convolution which is used to recover the size of segmentation figure in the image. So, after each up-convocation, it also has chain of feature that are with the same level. This is very helpful to find the segmented area from contraction path to expansion path. Finally, it converts features from 64 to 2, since the output contains only 2 grey level black and white.

Confusion Matrix is used for check performance of CNN Model for Classification and for Segmentation Evolution Jaccard index is used in U-net model. It calculates the overall pixel accuracy which measures the proportion of correctly classified pixels, however, it can be biased by imbalanced datasets. Mean single-class accuracy calculates the average proportion of correctly classified pixels in each class, which can also be biased by imbalanced datasets and overestimates the true accuracy due to combining multiple negative classes into one inference class.

Google Colab Environment is used for the training of both CNN and U-net Model.

4. Results

4.1. Classification of Pathological Myopia and Non-Pathological Myopia

In this paper, CNN with adam optimizer is used for pathological myopia and non-pathological myopia classification. The model achieved training and testing accuracy of 97.8% and 97.6% respectively as shown in Fig. 3.

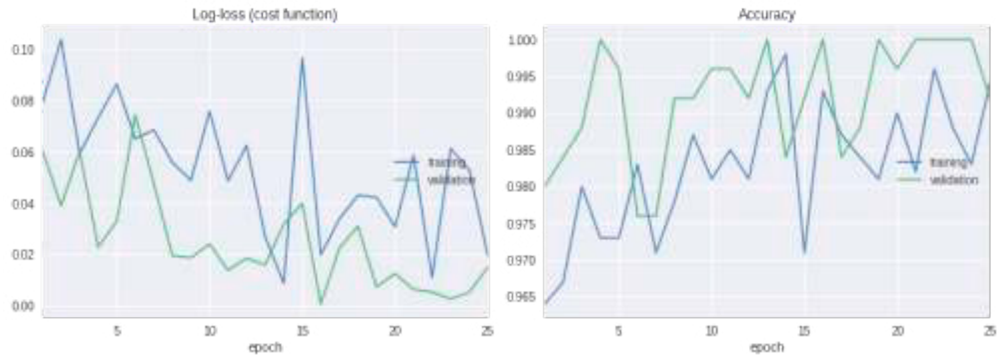

Fig. 3. Model Loss and Accuracy of CNN

4.2. Optic Disk Segmentation

The Parameters for CNN model with Adam Optimizer are: learning rate 0.001 and dropout 0.25. For optic disk segmentation, the model achieved the accuracy of Testing 96.7% and Training 98.9% which is shown in Fig.4.

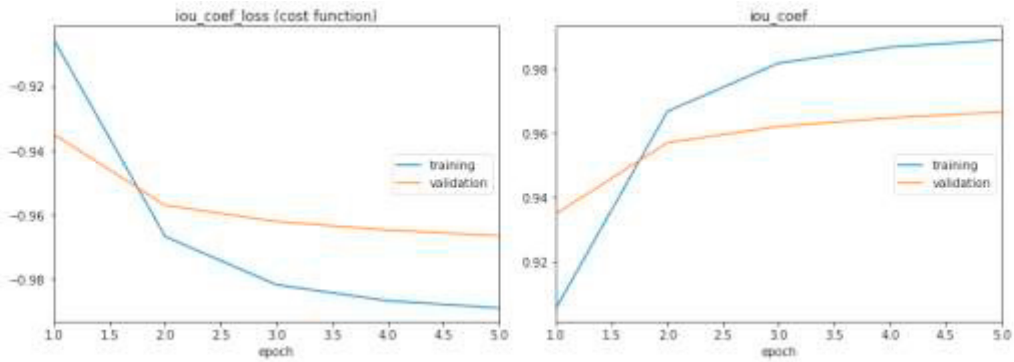

Fig. 4. Model Accuracy for Optic Disk Segmentation

4.3. Atrophy Lesions Segmentation

The parameters for atrophy lesions segmentation are same as the parameters used in section 4.2. The CNN model achieved the accuracy of Testing 96.0% and Training 97.9% which is shown in Fig.5.

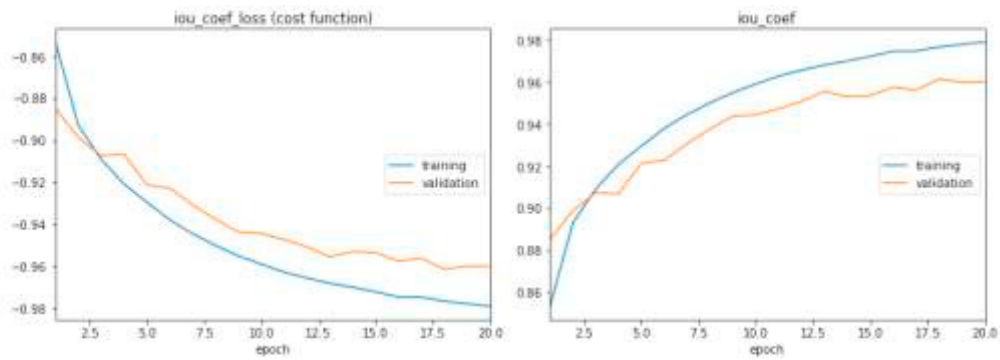

Fig. 5. Model Accuracy and Loss for Atrophy Segmentation

4.4. Result Comparison

Proposed method in this paper has achieved 97.8% accuracy for Classification of Pathological Myopia which is better result than Previous approach PAMELA that has achieved 87% with SVM approach.

5. Conclusion

In this paper, the classification of pathological myopia images and image segmentation of different lesions are done using deep learning technique, CNN and U-net method. The accuracy of the method is experimentally tested and it achieved good results. It achieves 97.8% accuracy when compared to PAMELA with SVM approach which has produced 87% accuracy. It is an effective method for the classification of Pathological Myopia images and segmentation of lesions for training samples. Data augmentation is used in this method to increase the robustness of the classifier. The reported results are superior to the automated analysis of pathological myopia images.

In future work, the method will be tested for more dataset of pathological myopia images for atrophy and retinal detachment segmentation.

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